

Multi-Height RTL2GDS Flow for Mixed Signal SoC

Week 1 Study Report — AI-Assisted Mixed-Signal Physical Design
Lokashsri M

Abstract— This report presents a Week 1 study of the open-source reference repository *vsdmixedsignalflow* [3], which demonstrates a complete mixed-signal RTL-to-GDS physical design flow integrating a 2:1 analog multiplexer (AMUX2_3V) with a digital SPI controller using OpenLane and the SkyWater SKY130A PDK [2]. The study employed an AI-assisted methodology — using Claude and ChatGPT — to understand the repository, decode LEF/LIB files, generate Docker setup commands, and debug flow errors. All AI-generated outputs were manually reviewed and verified before inclusion. This report covers the objective, a detailed repository summary, AI prompts and workflow, experiments attempted, results and observations, key challenges, and planned next steps.

Index Terms—mixed-signal SoC, OpenLane, SKY130, RTL-to-GDS, LEF, LIB, analog macro integration, physical design, AI-assisted EDA, OpenROAD

I. OBJECTIVE

This repository demonstrates a complete mixed-signal RTL-to-GDS physical design flow using open-source tools. It explains how an analog IP is integrated with digital logic using the SKY130 PDK and the OpenLane/OpenROAD flow to generate the final chip layout (GDSII). The project specifically addresses how the Place-and-Route (PnR) of a 2:1 analog multiplexer IP is carried out using OpenLane, and how existing IP layouts must be modified to ensure acceptance by open-source EDA tools.

II. REPOSITORY SUMMARY

Mixed-Signal SoC

A mixed-signal System-on-Chip (SoC) integrates both analog and digital circuits on a single chip. Digital modules are typically designed using RTL, while analog macros such as DACs, PLLs, or multiplexers are manually designed and then integrated into the digital layout during physical design. As process nodes shrink, this integration becomes increasingly complex, and a divide-and-conquer methodology is adopted — analog and digital structures are designed separately and combined during the PnR stage [3].

RTL-to-GDS Flow

The RTL-to-GDSII flow converts a digital hardware description written in Verilog into a manufacturable chip layout. The flow includes synthesis, floorplanning, placement, power distribution network (PDN) generation, routing, design rule verification, and final GDSII generation using open-source EDA tools. For mixed-signal designs, analog macros are injected into this flow via their LEF and LIB representations.

The OpenROAD Project

OpenROAD [5] is an open-source physical design tool that automates the complete backend VLSI design flow. It performs placement, clock tree synthesis, routing, timing optimization, and verification, enabling an end-to-end RTL-to-GDSII implementation without proprietary tools. OpenROAD serves as the core engine within the OpenLane framework used in this study

ITEM	PURPOSE
verilog	RTL design (top-level + behavioral model)
LEF	Physical abstraction of analog IP for PnR
LIB	Liberty timing library for STA
GDS	Final silicon layout output
OpenLane	Automated RTL-to-GDS flow framework
openROAD	Physical design engine (placement, routing)
SKY130	Open-source 130 nm process design kit
Magic	Layout tool to extract and fix LEF/GDS

Inputs Required: The physical design flow requires several carefully prepared input files. Table I summarises the key inputs and their roles.

Obtaining IP and Verilog Files: Analog Intellectual Property (IP) blocks are pre-designed functional modules reused in larger designs. The analog multiplexer (AMUX2_3V) used in this study was originally designed for the OSU018 technology node and was modified for SKY130 using the Magic layout tool with sky130A.tech. The Verilog behavioral model was obtained from the Efabless PicoRV-32 repository, and pin names were aligned with those in the LEF file before generating the LIB using the `verilog_to_lib.pl` Perl script [3].

LEF File: The Library Exchange Format (LEF) file provides an abstract physical description of the analog macro, including its dimensions, pin locations, routing blockages, and placement information.

LEF Limitations Found: The auto-generated LEF from Magic was missing pin declarations and CLASS, ORIGIN, SITE, SIZE fields. The macro height was not a multiple of the SKY130 site height (2.72 μm), blocking placement. Fixes applied via **Magic Tcl:** `port make`, `property LEFclass CORE`, `property LEFsite nithdbl`, `property FIXED_BBOX {0 24 874 568}`.

III. OPENLANE FLOW

The hardened analog macro requires an *interactive* flow. The complete 10-stage sequence is shown in Fig. 1.

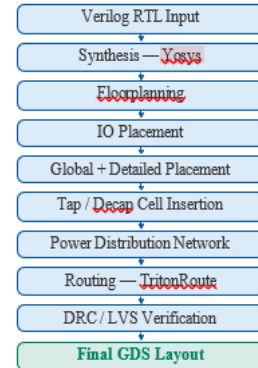


Fig. 1. RTL-to-GDS flow for the mixed-signal SoC.

Floorplanning

Floorplanning determines the chip dimensions and the locations of major design blocks, including analog macros. In this design, the AMUX2_3V macro is placed during the floorplan stage, and its position directly affects routing efficiency, timing, and utilisation. The command `init_floorplan_or` initialises the floorplan in OpenROAD.

Placement

Placement arranges all standard cells within the floorplan while satisfying timing and congestion constraints. Correct placement minimises wire length and improves circuit performance. In this flow, global placement (`global_placement_or`), detailed placement (`detailed_placement`), and tap/decap cell insertion (`tap_decap_or`) are performed sequentially.

Routing and DRC

Routing creates the metal interconnection between placed cells and macros. The command `run_routing` invokes TritonRoute within OpenROAD. Following routing, `run_magic_drc` verifies that the generated layout satisfies all semiconductor manufacturing rules defined by the SKY130 process technology. Any violations must be corrected before fabrication.

Final GDS and Flow Diagram

The final GDSII file is generated by `run_magic` and represents the complete physical layout of the integrated circuit, ready for foundry submission. The complete flow proceeds from Verilog RTL through synthesis, floorplanning, IO placement, global and detailed placement, tap/decap insertion, power distribution network generation, routing, DRC/LVS verification, and finally the GDS layout, as illustrated in Fig. 1.

IV. AI PROMPTS AND WORKFLOW

A *prompt* → *verify* → *apply* loop was followed throughout. All AI outputs were cross-checked against the actual repository.

- **P1** - “Explain the folder structure of `vsdmixedsignalflow` and its in-puts/outputs.”
- **P2** - “Explain MACRO, PIN, TIMING sections in this LEF. How does OpenLane use them?”
- **P3** - “Give Docker commands to install OpenLane on Ubuntu 22 and verify with `spm`.”
- **P4** - “LEF from Magic is missing CLASS and pins. What Magic Tcl commands fix this?”
- **P5** - “OpenLane: ‘Cannot find LEF macro for `AMUX2_3V`’. Cause and fix in `config.tcl`?”
- **P6** - “Generate a 1-page IEEE two-column LaTeX report for this Week 1 study.”

V. EXPERIMENTS ATTEMPTED

Tool setup: OpenLane installed via Docker [1]; confirmed working with the `flow.tcl` banner.

Repo inspection: All files (`config.tcl`, `script.tcl`, LEF, LIB, Verilog) cloned and manually reviewed.

LEF study: port make workflow for metal-layer labels and the poly-to-locali re-routing procedure for polysilicon-layer labels documented.

LIB generation: `perl verilog_to_lib.pl AMUX2_3V AMUX2_3V` verified as the correct command.

Flow execution: `run_synthesis` completed successfully (Yosys, no errors). Floorplan raised a height-mismatch warning. Full placement was not completed this week.

VI. RESULTS AND OBSERVATIONS

- ✓ Yosys synthesis produced a valid gate-level netlist without errors.
- ✓ The corrected LEF was loaded by OpenLane; all six pins (A, B, S, X, VPWR, VGND) were resolved correctly.
- ✓ AI assistance reduced LEF/LIB comprehension and Docker setup time by an estimated 2–3 hours.
- ✗ Macro height mismatch blocked placement legalisation.
- ✗ No GDS produced — flow stalled at detailed placement.

VII. CHALLENGES

- Incomplete LEF: Auto-generated LEF from Magic lacked pin declarations, CLASS, ORIGIN, SITE, and SIZE — all requiring manual Tcl commands to correct.

- Macro height mismatch: Height must be exactly 2.72 μm or 5.44 μm for the SKY130 `fd_sc_hd` standard-cell library.
- Poly-layer pin conversion: Polysilicon labels cannot be directly ported; re-routing through locali is required, adding manual steps.
- Config key deprecation: Several `config.tcl` keys from the repo are deprecated in the current OpenLane Docker image.
- Time constraints: A full synthesis-to-DRC run requires several hours; Week 1 focused on understanding and partial execution.

VIII. NEXT STEPS

- Fix macro height to 5.44 μm in Magic and regenerate a fully compliant LEF.
- Update `config.tcl` for OpenLane v2 key naming.
- Complete the flow through `run_routing` and obtain a routed DEF.
- Run `run_magic_drc` and document violations.
- View and screenshot the final GDS in KLayout.
- Add custom LIB timing constraints for the macro.
- Use Codex/Copilot to auto-generate Tcl fix scripts for recurring DRC errors.

IX. CONCLUSION

This Week 1 study confirmed that an AI-assisted workflow significantly accelerates the learning curve for mixed-signal physical design using open-source EDA tools. The `vsdmixedsignalflow` repository provides a comprehensive reference covering not only the OpenLane execution steps but also the critical pre-processing required to make an analog IP layout acceptable to PnR tools — a detail rarely covered in standard EDA tutorials. Although a complete GDS was not produced this week, the repository structure, LEF/LIB semantics, Magic-based pin-fixing procedure, and all flow stages are now thoroughly understood. Resolving the macro height mismatch and completing the placement-to-GDS stages remain the primary objectives for Week 2.

REFERENCES

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